

ITS - Industrial Technology Solutions

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CNC Plasma Cutting Machine - Service Report

Client:	SOMIK Skikda - Skikda
Date:	14/07/2026
Machine:	KOTEC KTPG-320 CNC Plasma Cutting Machine
Serial No:	149P324050WYPV1K

Machine Specifications

Parameter	Details
Manufacturer	KumKwang Tech Co., Ltd. (KOTEC) - Korea
Model	KTPG - 320
Date of Manufacture	November 2010
Controller	Sentrol 2 (SENTROL)
Servo Motors	YE LI YBL13S-75L.Z (1kW, 2000 RPM)
Servo Drives	YPV Series AC Servo Drives
Plasma System	Hypertherm HyPerformance Plasma

Part 1: Movements Control

Invoice No: ITS-2026-001

Problem: The machine has no PLC (Programmable Logic Controller). The Sentrol 2 CNC controller relies on electromagnetic relays for I/O, limiting motion control capabilities.

Work Performed:

1. Installed Mach3 software on dedicated PC All-in-One
2. Installed UC300 Motion Controller and connected to Mach3
3. Configured YE LI YBL13S-75L.Z servo motors (X, Y axes) with YPV Series servo drives
4. Installed DM556 digital stepper driver for the retained SLO-SYN Z-axis stepper motor

5. Wired the EPC encoder of the Z-axis into the UC300 fast inputs (MPG port)
6. Adapted UC300 with servo drives
7. Tested the first motor
8. Rewired all schematic control for the new system
9. Customized Mach3 interface
10. Configured travel limits and home positions
11. Tested full system
12. Movement tests and operation validation

Parts / Materials Used:

No.	Description	Qty	Local Price (DA)	Import Price (EUR)	Total
1	PC All-in-One	1	40,000	-	40,000 DA
2	Mach3 License	1	-	165 - 189	165 - 189 EUR
3	UC300 Motion Controller (300KHz, up to 6 axes)	1	-	89 - 150	89 - 150 EUR
4	DM556 Digital Stepper Driver	1	3,000	-	3,000 DA
5	Box Panel	1	5,000	-	5,000 DA

Part 1 Total: 403,000.00 DA

Part 2: Plasma Out

Invoice No: ITS-2026-002

Problem: The plasma system components exist at the machine but need to be integrated and configured with the new Mach3 CNC control system.

Work Performed:

1. Integrated Hypertherm HyPerformance Plasma system with Mach3 CNC
2. Installed standalone HF-rated Torch Height Control (THC) unit (Proma Compact THC SD) as an electrical barrier protecting the UC300 controller

3. Wired THC Step/Dir chain: UC300 -> THC unit -> DM556 -> Z-axis stepper motor
4. Configured Ignition Console (078172)
5. Configured Manual Gas Mix Console (078532)
6. Set up Gas Control and Metering Panel (O2/Air/N2)
7. Connected plasma torch and torch leads
8. Configured HF grounding and shielding of the ignition console (17 kV)
9. Configured safety interlocks (coolant, pressure, voltage)
10. Cutting tests and operation validation

Components Already at Machine:

No.	Component	Part No.	Status
1	HyPerformance Plasma Power Supply (HPR)	-	We Have
2	Ignition Console	078172	We Have
3	Manual Gas Mix Console	078532	We Have
4	Plasma Torch and Torch Leads	-	We Have
5	Gas Control and Metering Panel	-	We Have
6	Work Cable (Grounding)	-	We Have

Parts / Materials Used:

No.	Description	Qty	Local Price (DA)	Import Price (EUR)	Total
1	THC Unit (Proma Compact THC SD, HF-rated)	1	-	180 - 250	180 - 250 EUR

Part 2 Total: 150,000.00 DA

Summary

Invoice	Service	Amount
ITS-2026-001	Movements Control	403,000.00 DA
ITS-2026-002	Plasma Out	150,000.00 DA

TOTAL: 553,000.00 DA

Note: Imported parts (Mach3 License, UC300 Motion Controller, THC Unit) are priced separately in EUR.

For ITS

Signature and Stamp